Responsible AI Risk Assessment: Measuring and Mitigating Racial Bias in Healthcare Algorithms

Executive Summary

This paper investigates racial bias in a widely used healthcare algorithm, Optum. The algorithm helped determine which patients need additional attention and was found to favor white patients over black patients.[1]Hospitals across the United States have used the system developed by Optum, thereby perpetuating racial disparities by significantly underestimating the needs of black patients, who have more chronic health conditions compared to white patients.[2] Close to 200 million people were impacted by similar tools that are used by hospital networks and health-care systems nationwide.[3]

This analysis shows how NIST AI RMF could have prevented the Optum bias affecting 200M patients. The paper also explains how the algorithm allocates access to high-risk care management systems and quantifies its racial fairness problems using metrics such as demographic parity and equal opportunity, diagnosing why these problems arise.

The decisions and stakeholders impacted by this algorithm include:

1) Decision: Whether to continue using, and how to redesign, a health-risk algorithm that allocates access to high-risk care management programs for patients with chronic health conditions.
2) Stakeholders: Black and White patients enrolled in risk-based contracts, clinicians and doctors who rely on risk scores to make recommendations, hospital leadership who deploy the tool, and the algorithm vendor.[4]

The fairness targets that this paper looks at include:

1) Demographic Parity (DP) target: Increasing the share of black patients that are identified by the algorithm for care access. This ensures that the selection rate is comparable to that of white patients.
2) Equal Opportunity (EO) target: For patients with similar health-care needs, ensure similar true positive referral rates into care management systems, because the program's benefit should depend on clinical need rather than on racial or historical spending indicators.

The targets are measured using the following metrics:

For the DP target, only 17.7% of patients who are auto-identified for care management are black patients, as opposed to the 46.5% of white patients who were auto-identified. The DP ratio in this case is: 17.7 / 46.5 = 0.38

The EO target has a gap of about 15%, which means that among the top 15% of highest-need patients, black patients are significantly less likely to be flagged for the program, as they need to be considerably sicker than white patients to reach the risk-score required for flagging.[5]

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[1] Gawronski, Quinn. "Racial bias found in widely used health care algorithm." *NBC News*, 6 November 2019, https://www.nbcnews.com/news/nbcblk/racial-bias-found-widely-used-health-care-algorithm-n1076436.

[2] Ledford, Heidi. "Millions Affected by Racial Bias in Algorithm." 31 October 2019.

[3] Ledford, Heidi. "Millions Affected by Racial Bias in Algorithm." 31 October 2019.

[4] Paul, Kari. "Healthcare algorithm used across America has dramatic racial biases." *The Guardian*, 25 October 2019, https://www.theguardian.com/society/2019/oct/25/healthcare-algorithm-racial-biases-optum.

[5] Ledford, Heidi. "Millions Affected by Racial Bias in Algorithm." 31 October 2019.

The top mitigation strategy for this problem is to reframe the training label from future healthcare costs to a more direct measure of healthcare need. Instead of predicting only future cost, a new composite label that jointly captures health status and cost effectively redefines 'risk' in terms of clinical need rather than spending alone.

SECTION 1: GOVERN

This section covers risk management principles that address the full model lifecycle and associated processes, including regulatory, legal, and organizational policies the model must adhere to.[6]

GOVERN 1: Policies and procedures related to the MAP, MEASURE, and MANAGE functions of the AI system

Subcategory	RMF Expectation
Govern 1.1 Legal and regulatory requirement	The hospital and vendor understand which healthcare, civil-rights, privacy, and anti-discrimination laws apply when using an AI system to allocate access to care-management programs, and they align their AI risk-management practices with those requirements. That means documenting how the model is tested for disparate impact across protected groups (e.g., race), how patient data are handled under HIPAA, and how transparency obligations to patients will be met. The organization maintains written policies and regular training so that clinicians, data scientists, compliance officers, and leadership know the legal boundaries for using demographic data and how to respond when bias is detected.
Govern 1.2 Trustworthy AI Characteristics	Trustworthy characteristics, including transparency, fairness, reliability, accountability, and validity, must be integrated into the AI policies and procedures for organizational risk management. This includes detailed standards for risk mapping, model training, testing, and validation, as well as for aligning these policies with best industry practices and legal standards.[7]
Govern 1.3 Procedures for risk management	For this model, GOVERN 1.3 would include establishing a risk-tiering system: (Red-Amber-Green) based on impact × likelihood: Red (Critical): DP ratio <0.5 or EO gap >0.05 → requires immediate deactivation Amber (High): DP 0.5-0.8 or EO 0.02-0.05 → enhanced monitoring, subgroup-adjusted thresholds. Green (Acceptable): DP ≥0.8 and EO ≤0.02 → routine quarterly monitoring. Policies would specify:

[6] National Institute of Standards and Technology. *Artificial Intelligence Risk Management Framework (AI RMF 1.0).* NIST AI 100-1. Washington, DC: U.S. Department of Commerce, 2023. https://doi.org/10.6028/NIST.AI.100-1

[7] National Institute of Standards and Technology. *Artificial Intelligence Risk Management Framework, 23*

	• Impact assessment: Quantify the number of high-need Black patients missed. • Likelihood tracking: Monitor enrollment drift by race through automated dashboards. • Documentation: Risk scores (impact × likelihood) logged in centralized model-risk repository with RAG status, mitigation timelines, and responsible parties.[8]
Govern 1.4 Documentation of Risk-Management	AI Risk management policies must have clear documentation covering vendor and hospital network contact information, use cases, scope, risks, and limitations of the model, stakeholder engagement, auditing, model training, and validation plans.
Govern 1.5 Organization roles & continuous monitoring	Plans for continuous monitoring of the model, along with policies defining organizational functions and personnel responsible for AI monitoring, should be established within incident response plans to enable real-time flagging of potential incidents.
Govern 1.6 AI Inventory Systems	Establishing an inventory system for the algorithm that acts as a repository of information on the system documentation, incident response plans, source code, and relevant AI actors, allowing for the resolution of high-level queries.[9]
Govern 1.7 Decommissioning Procedures	When fairness metrics exceed risk tolerances, execute safe decommissioning: (1) Immediate containment – pause auto-enrollment at 97th percentile, route all 55th+ percentile patients to manual clinician review; (2) Patient impact mitigation; (3) Communication—notify impacted communities via patient portals; (4) Reputational management—public statement acknowledging the issue and mitigation steps taken; (6) Establishing a clear transition plan.

GOVERN 2: Accountability Structures related to MAP, MEASURE, and MANAGE functions

Subcategory	RMF expectation
Govern 2.1 Roles & Responsibilites	Clear roles related to AI risk management must be delineated, including the board of directors, auditors, evaluators, designers, managers, and oversight. Through GOVERN 2.1, clear lines of communication must be established among all AI Actors (clinicians, patients, senior management in the hospital network, and vendors), and policies must be implemented to incentivize AI

[8] National Institute of Standards and Technology. *NIST AI RMF Playbook*. Gaithersburg, MD: National Institute of Standards and Technology, 2023. https://www.nist.gov/itl/ai-risk-management-framework/nist-ai-rmf-playbook, 7

[9] National Institute of Standards and Technology. *NIST AI RMF Playbook*, 8

	Actors to act in accordance with industry best practices and the hospital network's existing compliance and legal policies.[10]
Govern 2.2 AI Risk Management Training	All AI actors must receive training in AI risk management to enhance adoption across organizations; this includes training in interpreting model results, potential negative risks, escalation criteria, trustworthy AI characteristics, and organizational policies on the use of the healthcare algorithm.
Govern 2.3 Executive leadership	Executive leadership and senior-level leadership of healthcare networks should maintain final responsibility and authority over managing risks related to the model. This includes delegating resources to perform risk management, declaring risk tolerances (DP and EO > set thresholds), and actively supporting risk management efforts as part of the organizational culture.

GOVERN 3: Workforce diversity processes related to the MAP, MEASURE, and MANAGE functions

Subcategory	RMF Expectation
Govern 3.1 A diverse team guides decision-making	Define policies and hiring practices to ensure a diverse body working across the healthcare algorithm. This includes community representatives, data scientists, clinicians, equity specialists, and healthcare analysts to ensure sufficient domain expertise to address any emerging socio-technical risks arising from the model's use.[11]
Govern 3.2 Human-AI roles	For this model, policies would clearly define: AI System Overseers (Governance): 1) AI Review Board (a diverse body) approves deployment, sets risk tolerances, and reviews quarterly fairness reports. 2) Vendor liaison monitors third-party updates and subgroup performance data. 3) Analysts run automated DP/EO monitoring dashboards. AI System Operators/Users: > Primary care clinicians receive risk scores at the 55th percentile (97th percentile auto-enrolls), and document overrides with clinical rationale. > Care-management coordinators execute enrollment decisions and track patient outcomes by race.

[10] National Institute of Standards and Technology. *NIST AI RMF Playbook*, 16
[11] ibid, 19

GOVERN 4: Fostering a culture that considers and communicates AI risks

Subcategory	RMF Expectation
Govern 4.1 Safety-first mindset policies	For this model, policies to cultivate critical thinking include: Three Lines of Defense: 1st Line (Clinicians/Data Team) flags biased risk scores during use, 2nd Line (Equity/Risk Officers) runs mandatory DP/EO audits pre-deployment, 3rd Line (External Audit) independently validates vendor claims and fairness metrics Effective Challenge Mechanisms include veto power over deployment if equity powers fail to meet thresholds, halting auto-enrollment if systemic gaps are identified, and red-teaming, which includes testing the model under stress conditions to identify vulnerabilities.[12]
Govern 4.2 Documents risks and impacts	Conducting iterative impact assessments to frame risks related to the use of the healthcare tool; these assessments must include feedback and input from relevant stakeholders, such as clinicians and patients, to evaluate the model and assist in decisions on deployment or decommissioning.
Govern 4.3 Testing, incident identification, and information sharing	For this healthcare model, practices include: Mandatory Testing Protocols: Pre-deployment fairness stress tests, and red-teaming with high-need Black patient edge cases. Incident Detection Pipeline: Automated alerts for fairness violations, clinician override spike detection (>20% override rate). Information Sharing practices include logging all incidents to the inventory system, including model details, affected demographics, and remediation steps. Additionally, hospital networks impacted by severe racial biases can share lessons learned with other healthcare networks using similar models.

GOVERN 5: Engagement processes with AI Actors

Subcategory	RMF Expectation
Govern 5.1 Integrating external feedback	Policies and procedures to be established by the hospital network to collect and effectively integrate external feedback from independent AI evaluators and end-users, particularly regarding user interaction, minimizing errors, and the model's negative impacts.
Govern 5.2 Incorporating feedback	Policies can look like a feedback validation process in which a cross-functional committee meets quarterly to review external input from qualitative and quantitative data, including metric reviews and surveys. Feedback is scored by relevance and

[12] ibid, 23

from AI actors	impact using RAG scales. Stakeholder input directly adjusts DP/EO thresholds. Iteratively conduct impact assessment via RAG, where each feedback item is assessed for impact severity x likelihood.

GOVERN 6: Risks from third-party software

Subcategory	RMF Expectation
Govern 6.1 Third-party AI policies	Policies require vendor contracts to mandate: (1) transparency on training data provenance, label methodology, and subgroup performance (DP/EO by race), and (2) right-to-audit model internals and retraining processes.
Govern 6.2 Contingency processes	Establish policies to address the vendor's model failures, including but not limited to transparency, bias, and privacy risks. Ensure that appropriate incident-response plans established in GOVERN 1.3 address risks posed by third-party software.[13]

SECTION 2: MAP

The map function establishes the context to frame risks related to the healthcare algorithm. This section describes the context, stakeholders involved, and the risk profile of this algorithm, setting the stage to assess metrics and fixes.

MAP 1: Establishing context

Subcategory	RMF Expectation	Gaps in the model (with evidence)
Map 1.1 Context Setting	Intended Purpose: To identify patients with chronic health conditions, and recommend them to high-risk care-management systems Potential Benefits: Helps identify high-risk patients and improve outcomes for these patients by enrolling them in intensive healthcare systems. By predicting future utilization, it supports the hospitals in planning staffing and budgets. Negative Impacts of the System: Although the model was not designed to be racially biased, the incorrect use of labels led to significant racial bias in the outcomes. Assumptions and limitations:	In deployment, the hospital and vendor used the model's future-cost score directly to prioritize patients for high-risk care-management, without explicitly documenting or challenging the assumption that cost is an adequate proxy for health need. The system excluded race as an input feature. Still, no pre-deployment or early-deployment analysis was conducted to check for racial performance differences or to validate that EHR/claims data captured illness burden equitably

[13] ibid, 32

	The system is designed to predict future healthcare costs, and the cost, as a label, is used as a proxy for health needs. According to the model: ● Higher spending indicates greater need. ● Excluding race from the feature set is sufficient to avoid racial bias. ● EHR data captures illness burden across all racial groups.[14]	across groups. Over time, it became clear that the model was racially biased against black patients, leading to lower enrollment rates in care-management programs.
Map 1.2 Interdisciplinary Collaboration	Mapping the context of a system requires a team of AI actors with diverse experiences, backgrounds, and skill sets. This enables a better understanding of the tool's purpose and function through collective perspectives.	The original development and deployment were largely vendor and health-system-analyst-driven, with little to no input from clinicians and community stakeholders. Post-deployment, once the biases were identified, collaboration among academicians, scientists, clinicians, and the vendor team was observed to re-analyze the model.[15]
Map 1.3 Organization's Mission	The mission was to identify high-risk patients and provide them with extra care and management to improve overall health outcomes. The goal was to accurately identify patients who need enrollment in such systems.	Large healthcare providers and systems use these algorithms to accurately and efficiently identify high-risk patients, rather than manually identifying and assigning them to care-management systems. These algorithms are widely considered effective at improving outcomes at a low cost, as hospitals often incur substantial expenses to support such programs.
Map 1.4 Business Value	The organization explicitly defines and periodically re-evaluates the business value	For this tool, the business value was framed as cost prediction and

[14] Ledford, Heidi. "Millions Affected by Racial Bias in Algorithm
[15] ibid, 5

	and use context of the AI system, including: what decision(s) it supports (e.g., enrollment into high-risk care-management), how success is measured (clinical outcomes, equity, cost), which populations are in-scope, and how trade-offs between efficiency and fairness will be handled.	population health efficiency, without clear documentation of a commitment to equity and fairness for all racial groups. Only after an external study emerged did the organization begin to question whether future costs would align with its mission and business value.[16]
Map 1.5 Risk Tolerance	Risk tolerance reflects the extent to which vendors and healthcare systems are willing to take on risk in carrying out their missions. In healthcare, this should include explicit limits on disparities in access to care, compliance with regulations and professional guidelines, and clear thresholds for when predictive bias becomes unacceptable in a given use case.[17]	No risk tolerances were documented before the model's deployment. The significant disparities in outcomes across groups suggest that equity-related risks were considered acceptable until external studies exposed the problem.
Map 1.6 System Requirements	Ensuring that the system requirements of the prediction tool are clearly documented, keeping in mind the tool's end-users and the societal implications of deploying it. This enhances the AI model's trustworthiness.	No documentation of system requirements was done, keeping in mind metrics related to fairness, equity, and societal implications of deployment.

[16] Ibid, 5

[17] National Institute of Standards and Technology. *NIST AI RMF Playbook*, 66

Recommendations: To strengthen MAP 1 (Context Establishment) for this model, the following actions are recommended for future deployment:

- A clear document of assumptions and limitations, including evidence for where proxy labels can fail for different groups.
- Building interdisciplinary teams for future testing and deployment that reflect broad domain experience.
- Define success metrics that include both efficiency and fairness (e.g., EO/DP targets for enrollment), and require periodic review of whether the current label and model still serve those goals.

MAP 2: Categorizing the AI System

Subcategory	RMF Expectation	Gaps in the model (with evidence)
Map 2.1 Specific Task	The model is a predictive risk-scoring tool that uses prior-year claims and EHR data to generate risk scores that prioritize chronically ill patients for enrollment in care-management systems.	The model is a predictive risk-scoring tool that uses prior-year claims and EHR data to generate risk scores. Clinicians assess these risk-scores, and based on their recommendations, chronically ill patients are enrolled into care management systems.
Map 2.2 Knowledge Limits	Information about the system's knowledge limits, including bias, fairness, and relevant assumptions based on the proxy label, is clearly documented, along with guidelines for its use by AI actors.	There is no documented evidence on the knowledge limits of this model during deployment. The guidelines for human use included clinicians in the loop and using their recommendations for the final say on enrollment. However, clinicians' recommendations were mainly based on the risk-scores given in the model, leading to biases.
Map 2.3 Scientific Integrity	Testing and establishing evaluation protocols to confirm that the model is operating as designed. Key metrics, including clinical utility, calibration by race, DP, TPR, and stakeholder trust, are assessed through model testing and evaluation. TEVV considerations are identified to ensure the model is	No evidence of documentation of any of the integrity considerations mentioned.

	trustworthy and reliable.[18]	

Recommendations: To strengthen MAP 2: Categorization for this algorithm, the following actions are recommended for future deployment:

- Develop and apply TEVV protocols to the healthcare model
- Ensure the incorporation of metrics related to bias and fairness, and ensure they are interpretable for any downstream decision-making taking into account socio-technical factors that can cause biases.

MAP 3: AI Capabilities, usage, and goals

Subcategory	RMF Expectation	Gaps in the model (with evidence)
Map 3.1 Potential Benefits	The model improves outcomes for chronically ill patients by accurately identifying candidates for high-risk care-management programs, while reducing hospital staffing/budgeting costs and preventing high-cost utilization (e.g., hospitalizations). Processes of regular communications should be established with end-users.	The model improved outcomes for white patients; however, it was not bias-free, and therefore, equitable access was ignored as a core value, prioritizing cost prediction/efficiency. No communication processes were formally set up with the affected group.
Map 3.2 Potential Costs	Potential costs from AI errors and poor trustworthiness are systematically examined and documented, including monetary costs (e.g., wasted care-management resources, preventable hospitalizations) and non-monetary costs (e.g., harm to patient health, reputational damage, legal/regulatory risks). These are explicitly linked to organizational risk tolerances and reviewed with stakeholders before deployment.	No documented analysis of potential costs from model errors existed before deployment. Monetary costs of under-enrolling sicker Black patients and non-monetary costs, such as eroding trust, were not anticipated or quantified.
Map 3.3 Application	The application scope is limited to insured users in the U.S., and it distinguishes only	This was a widely used model that impacted 200 million users annually

[18] National Institute of Standards and Technology. *NIST AI RMF Playbook*, 79.

scope	between Black and White patients.[19]	nationwide.
Map 3.4 Operator Proficiency	Processes are established to ensure clinicians and operators understand the AI system's performance limits, trustworthiness characteristics, error modes, and relevant clinical standards/certifications.	No evidence of pre-deployment clinician training on the model's reliance on cost-as-proxy, its racial performance gaps, or guidance for overriding biased risk scores. Clinicians received risk scores and patient context at the 55th percentile threshold, but lacked documentation of known limitations
Map 3.5 Human Oversight	Ensuring human-in-the-loop and rigorous oversight procedures before deploying an AI system.	No documented human oversight processes were in place for clinicians reviewing risk scores. While primary care physicians received patient context at the 55th percentile for enrollment consideration, there were no defined protocols for identifying/acting on racial disparities in risk scores, no override tracking, and no linkage to GOVERN policies.

Recommendations: The actions to strengthen MAP 3: AI Capabilities include:

- Developing training materials for clinicians about context of use, known limitations, and the system's performance
- Perform context analysis to accurately map the negative impacts of the model that arise when bias and fairness metrics are not included.
- Actively engage with end-users and establish feedback loops with clinicians and patients to better document limitations

MAP 4: Risks and Benefits

Subcategory	RMF Expectation	Gaps in the model (with evidence)
Map 4.1 Mapping	Processes are established and followed to map the technical risks of AI components and the legal risks (e.g.,	No documented mapping of third-party vendor risks occurred, including the legal exposure from deploying racially biased

[19] Ledford, Heidi. "Millions Affected by Racial Bias in Algorithm"

	third-party data/software, liability for biased outcomes, regulatory non-compliance), with clear mitigation steps and vendor risk assessments.	predictions in clinical decisions.
Map 4.1 Internal Risk Control	Establishing internal risk controls and developing evaluations for third-party technologies (the healthcare algorithm in this case) utilised, which may have privacy and bias concerns.[20]	No review of the algorithm by hospital networks was conducted before deployment to assess privacy and bias risks.

Recommendations: To strengthen MAP 4: Risk Mapping, the following actions are recommended:

- Conduct and review existing audit reports, testing results, and end-user license agreements related to the vendor's technology
- Review all material related to the algorithm including the data and the models for any risks related to bias and data privacy

MAP 5: Model Impacts

Subcategory	RMF Expectation	Gaps in the model (with evidence)
Map 5.1 Benefits & Harms	Vendor, clinicians, and hospital networks should evaluate and document the likelihood and magnitude of positive and negative impacts, based on expected use, prior use of such models, and stakeholder feedback. TEVV resources should be assigned based on the likelihood of such estimates.	Before deployment and even after, there is no evidence of impact assessments taken to assess the benefits and harms of the model. Only much later, after deployment, did Optum collaborate with researchers to retrain and reanalyze the model's feature set.
Map 5.2 Incorporating Feedback	Establishing and documenting engagement practices with all stakeholders, particularly the model's users, to ensure that limitations and impacts are accurately assessed.	The hospital networks and vendors did not proactively engage with the stakeholders to understand the model's impact. Lack of engagement practices exacerbated the issue, causing significant biases in outcomes.

[20] National Institute of Standards and Technology. *NIST AI RMF Playbook*, 86

Recommendations: To strengthen MAP 5: Assessing model impacts', the following actions are recommended:

- Apply TEVV regularly across all stages of the AI lifecycle — from vendor selection and pre-deployment testing to post-deployment monitoring — with documented results stratified by race and clinical need.
- Establish and document stakeholder engagement in the early stages of system design. (eg: cross-functional review boards). Additionally, identify and establish approaches to engage and incorporate input from clinicians and patients' to ensure continuous monitoring

LIFECYCLE HOT SPOTS

Block	Evidence
Data Labelling	Commercial risk-prediction algorithm generates risk scores. Patients above the 97th percentile are automatically identified, and those above the 55th percentile are referred to their primary care physician.[21]
Stakeholder & harms	Black and white primary-care patients, clinicians, hospital leadership, and vendors. Black patients receive systematically lower risk scores; only 17.7% of black patients are auto-enrolled, opposed to the 46.5% white patients enrolled.[22]
Data label and proxies	Future total medical costs are a proxy for health needs; black patients spend $1800 less on health, dropping their risk scores. [23]
Harm Hypotheses	Cost proxy underestimates the true health needs of black patients, causing Black TPR rates to be much lower than the White TPR rates
Fairness metrics (DP, EO)	DP ratio is 0.38, indicating significant disparities, as it falls well below the fairness benchmark (0.80 rule). EO also indicates a 15% gap in selection rates between clinically similar patients.
Lifecycle Hot Spots	The cost label contains spending disparities. In terms of training/thresholding, the model is global, with no subgroup evaluation, and is fixed at the 97th and 55th percentiles. In the deployment, the clinician does not correct for bias, and there are no fairness checks by race.

[21] Ledford, Heidi. "Millions Affected by Racial Bias in Algorithm
[22] ibid
[23] ibid

SECTION 3: MEASURE

This section analyzes and monitors AI risks and related impacts, which include thorough documentation of the system's functionality and trustworthiness.

NOTE: Unlike the MAP analysis, which provided detailed documentation of system context, deployment rules, and observed outcomes, there is no evidence in the readings or other reports of systematic measurement of fairness metrics, risk management processes, or governance oversight for this third-party algorithm. Given the proprietary nature of the vendor technology, the hospital appears to have lacked visibility into and control over core lifecycle stages beyond initial deployment. The following sections, therefore, articulate NIST AI RMF expectations for what should have been implemented across these critical functions to ensure trustworthy AI use in high-stakes healthcare decisions.

MEASURE 1: Methods and Metrics

Subcategory	RMF Expectation
Measure 1.1 Metrics	Select and document metrics that explain predictive performance (calibration, concentration of high-need patients) and fairness (DP, EO, FPR/FNR stratified by race) for all deployment thresholds (97th and 55th). These metrics should be calculated pre-deployment, with continuous monitoring in post-deployment.
Measure 1.2 Appropriateness of Metrics	The metrics listed above provide an accurate and useful measure of the performance. The healthcare networks implementing the model should create automated reports/dashboards to continuously monitor the metrics, with alerts when fairness violations exceed risk-tolerance limits.
Measure 1.3 Experts Involved	Utilize separate testing teams to test and analyze the model's results continuously; these teams should ideally be uninvolved in the model's front-line development. This team should have broad domain exposure, as detailed in Map 1.2, and regularly evaluate stakeholder processes and inputs, as detailed in Map 3.1.[24]

MEASURE 2: Measuring Trustworthiness

Measure 1.1 Documenting TEVV	Specific Metrics: DP, EO, FPR, FNR. Assessed through counterfactual simulations and threshold sensitivity analysis.
Measure 2.2 Human Subject	The analysis covered insured patients in the U.S. healthcare system and distinguished between only Black and White patients. In a real-world

[24] National Institute of Standards and Technology. *NIST AI RMF Playbook*, 96

Requirements	context, it is important to clearly identify the population for intended use and use disaggregated evaluation methods by race, age, gender, etc., to improve the algorithm for real-world use.
Measure 2.3 Managing risks	To effectively manage risks related to the model's impacts and the experience it delivers for end-users, it is important to test and evaluate the algorithm under non-optimal scenarios and document performance criteria for validity (FPR, FNR) and efficiency (prediction time and cost). In the model's case, it achieved high predictive power for cost but was not aligned with clinical need, biasing validity estimates.
Measure 2.4 AI System Functionality	Implement continuous monitoring for algorithmic drift using control limits on key metrics—DP ratio (target: ≥0.8), EO gap (target: ≤0.05), and calibration by race —with 95% confidence intervals. Track input drift in patient demographics/subgroup mix and outcome drift in cost/health proxies. Use error propagation analysis to quantify uncertainty in risk scores at 97th/55th percentile thresholds.
Measure 2.5 Validity and Reliability	The model was valid for its narrow technical objective: predicting future costs, but unreliable for the clinical deployment context. It was well-calibrated on costs across races, but systematically failed to identify high-need Black patients, showing poor generalizability. Document approaches to measure both internal and external validity, and clearly define the socio-technical context under which the AI system is set to operate.[25]
Measure 2.6 Safety Risks	The model violated safety risks by under-identifying high-need Black patients, and the harms caused exceeded the risk-tolerance limits set by benchmarks with DP > 0.80. In the post-deployment scenario, establish methods to stress-test the performance under scenarios such as concept drift and high load, and continuously consult with domain experts on the results.
Measure 2.7 Security and Resilience	A low-priority concern here is direct cyberattacks on the deployed risk scores themselves. Healthcare algorithms like this face greater risks from data poisoning during vendor training, biased label choice (already identified), or deployment misuse than from runtime security breaches. Resiliency tests to be established include stress-testing under adversarial conditions.
Measure 2.8 Transparency and	Maintaining audit logs to review sources of bias, error, and vulnerability in the algorithm. Establish policies for escalation regarding AI systems,

[25] National Institute of Standards and Technology. *NIST AI RMF Playbook*, 108

accountability	documenting go and no/go decisions made by AI actors.[26]
Measure 2.9 Interpretability	The algorithm model should be explained, validated, and documented so that clinicians and governance bodies understand what it predicts (future costs), how it was trained, and where it is unreliable for inferring true health needs. This includes model cards describing data sources, label choice, key assumptions (cost as proxy for need), and subgroup performance metrics (DP, EO, FPR, FNR by race). To support interpretability, local explanation methods such as LIME and SHAP should be applied to typical and high-stakes cases, showing which features drive individual patients' risk scores and whether these drivers differ systematically by race.
Measure 2.10 Privacy Risks	Ensuring the incorporation of privacy-related frameworks such as HIPAA, NIST AI RMF, and privacy by design principles to protect patient health information, manage AI risks, and incorporate privacy values into the design of the algorithm.
Measure 2.11 Fairness and Bias	Quantify the harms associated with the model using metrics such as DP, EO, and statistical hypothesis tests, and analyze these harms using impact assessments that use these metrics and inputs from affected communities.
Measure 2.12 Environmental Impacts	Energy and water consumption, GHG emissions, and resource consumption metrics can be used to assess environmental impact and sustainability assessments.[27]
Measure 2.13 Effectiveness of TEVV	Continuously review and assess metrics of EO and DP, along with accuracy and precision metrics for identifying and measuring risks in the algorithm.

Confusion Matrix Baseline – Calculations based on ("Optum_Toy_Dataset")

White Patients	**Referred=1**	**Referred=0**
Need=1	105	54
Need=0	6	414

Black Patients	**Referred=1**	**Referred=0**
Need=1	38	103
Need=0	1	27

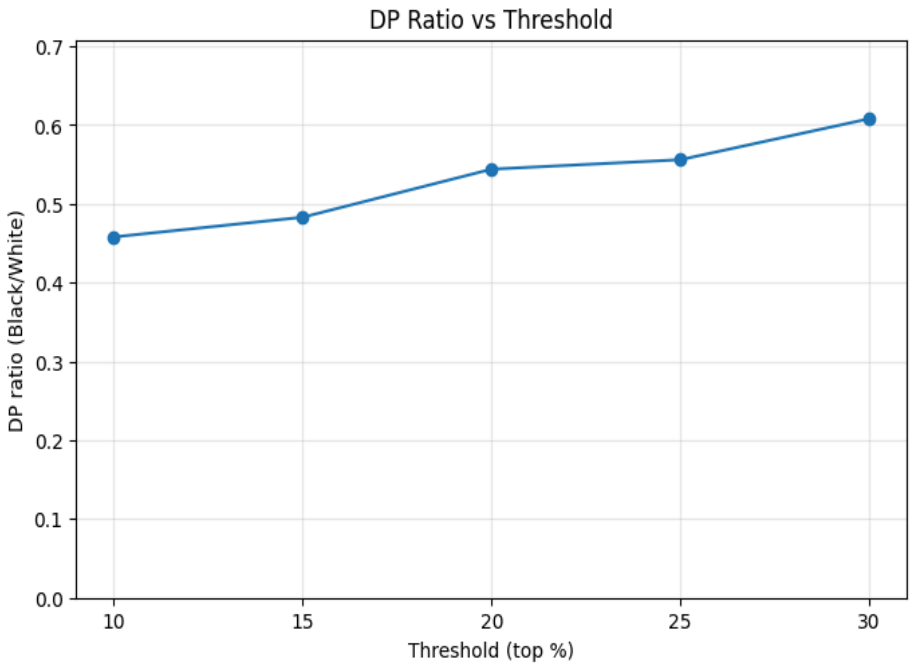
[26] National Institute of Standards and Technology. *NIST AI RMF Playbook*, 114
[27] ibid, 123

Group Fairness Metrics – Calculations based on (“Optum_Toy_Dataset”)

Metric	White	Black	Disparity ratio	Interpretation
Selection Rate (DP)	0.192	0.093	0.48	Black patients are referred at only about 48% of the White referral rate.
Equal Opportunity (TPR)	0.66	0.26	-0.39	White patients with actual need are identified at a much higher rate than Black patients.
False Positive Rate (FPR)	0.014	0.004	-0.01	White patients have a higher false positive rate than Black patients, so unnecessary referrals are slightly more common among White patients.
False Negative Rate (FNR)	0.34	0.73	0.39	The false negative rate is much higher for Black patients than for White patients. This means Black patients who actually need referral are substantially more likely to be missed by the system.

Threshold Sensitivity (From a dummy dataset on Optum)

Threshold (Top %)	DP Ratio	EO Gap	Notes
10	0.46	-0.28	At the 10% threshold, Black patients are selected at about 46% of the White selection rate, and Black patients have a substantially lower TPR than White patients
15	0.48	-0.39	At the 15% threshold, Black patients are selected at about 48% of the White selection rate, with a moderate improvement in the EO gap.
20	0.54	-0.40	At the 20% threshold, Black patients are selected at about 54% of the White selection rate, with a moderate improvement in the EO gap.
25	0.55	-0.38	At the 25% threshold, Black patients are selected at about 55% of the White selection rate, the EO gap reduces and remains negative
30	0.60	-0.30	At the 30% threshold, Black patients are selected at about 60% of the White selection rate. The EO gap stays negative at every threshold, meaning Black patients consistently have a lower true positive rate than White patients.



MEASURE 3: Tracking Risks in AI models

Subcategory	RMF Expectation
Measure 3.1 Emergent AI Risks	Dedicated personnel (AI risk officer, fairness analysts) and processes are established to identify and track risks across the AI lifecycle continuously. Develop a documented AI incident response plan triggered by fairness metric violations and establish cross-functional teams to quantify harms (Measure 2.11).
Measure 3.2 Hard to measure AI risks	For risks difficult to quantify with standard metrics (e.g., long-term health outcome disparities from under-treatment, trust erosion in marginalized communities), implement proxy tracking and qualitative monitoring: (1) Track proxy outcomes like emergency visit rates by race for enrolled vs non-enrolled patients; (2) Conduct quarterly patient satisfaction surveys stratified by race, asking about perceived fairness of care allocation.[28]
Measure 3.3 Feedback Processes	Analyze end-user requests and appeals in response to the model, and utilize feedback by systematically collecting and analyzing clinician overrides, patient outcomes, and qualitative stakeholder input to validate and refine fairness/reliability metrics.

[28] ibid, 131

MEASURE 4: Gathering Feedback

Subcategory	RMF Expectation
Measure 4.1 Measurement Approaches	Clinicians, hospital networks, and vendors should evaluate feedback from impacted communities through the processes established in Govern 5.1 and 5.2 and Map 1.6 and 5.1.[29]
Measure 4.2 Having domain experts	Measurement results (DP ratios, EO gaps, calibration curves by race) from deployment monitoring are regularly reviewed by domain experts, including clinicians, health equity specialists, data scientists, and patient advocates, to validate whether the system performs as intended across contexts.
Measure 4.3 Performance Changes	Develop a baseline quantitative measure for EO, DP, FPR, and FNR. Complement quantitative analysis with qualitative monitoring of clinician and patient feedback, performing sensitivity analysis to examine variances in performance with/without updates to the model.[30]

SECTION 4: MANAGE

This section analyzes the risks based on MAP and MEASURE assessments and entails allocating resources to establish risk treatments to respond to and communicate incidents effectively.

MANAGE 1: Examining Risks from MAP and MEASURE

Subcategory	RMF Expectation
Manage 1.1 Should the model be deployed?	Since the model was racially biased against high-need black patients, the algorithm has not achieved its intended purpose. The model should be deployed only after incorporating fairness metrics and utilizing the TEVV outputs from the MAP and MEASURE functions when examining performance.
Manage 1.2 Treatment of AI Risks	The key risk the model needs to treat is the under-identification of chronically ill black patients, and this must be prioritized for lower risk tolerance, receiving the greatest oversight and management resources.
Manage 1.3 Risk Response	For this healthcare risk-prediction model, a strong approach would include: (1) mitigating the risk by redesigning the label (health + cost or direct health measures), tightening fairness tolerances (e.g., EO $\lvert\Delta TPR\rvert \leq 0.02$). Procedures should prioritize harms that affect clinical safety and equity – missed enrollment of sicker Black patients.

[29] ibid, 134
[30] ibid, 140

Manage 1.4 Unmitigated Risks	Unmitigated or residual risks from improperly resolving model issues (e.g., continued racial bias) should be documented and communicated through procedures that disclose these risks to relevant AI actors. Making these residual risks transparent allows future purchasers and clinicians to weigh the algorithm's clinical benefits against remaining fairness and safety concerns.

MANAGE 2: Strategies to maximize AI's positive impacts

Subcategory	RMF Expectation
Manage 2.1 Resources for managing resources	These steps involve establishing risk management practices in accordance with the risk tolerance limits created. ● Black share among auto-enrolled high-risk patients to be at least 0.8 of the unbiased target in the 97th percentile (DP) ● The absolute difference in true-positive enrollment rates between Black and white patients must satisfy $\lvert\Delta TPR\rvert \leq 0.02$; anything larger is intolerable. Continuous monitoring of these metrics with external expertise to manage high risks.
Manage 2.2 Value sustenance	Establish clear risk controls that encompass trustworthiness characteristics, including algorithmic bias assessments, TEVV protocols, and mechanisms to avoid past failed designs involving racial bias. This includes label-design guardrails, interdisciplinary design reviews, and an internal catalog of past failures for accurate bias tracking.[31]
Manage 2.3 Unknown risk management	Continuous monitoring of the prediction model to detect new and unknown risks includes establishing procedures for detecting drift and decontextualization. Additionally, feedback loops and open lines of communication are established with all end-users to capture potential negative impacts and enable mitigation.
Manage 2.4 Mechanisms to deactivate the model	If the model repeatedly exceeds the established risk-tolerance thresholds, mechanisms must be in place to safely deactivate it within hospital networks. Materials from the functioning model should be preserved for legal review, and an internal root cause analysis can be conducted upon deactivation to understand why thresholds were exceeded and inform redesign and future safeguards.[32]

MANAGE 3: Managing third-party entities

Subcategory	RMF Expectation
Manage 3.1 Risks and	Data quality from the third-party vendor is consistently tested for drift and bias issues. Hospital networks should establish processes for third-party vendors to report potential

[31] ibid, 52
[32] ibid, 53

benefits from third-party	vulnerabilities and biases in the model prediction.
Manage 3.2 Monitoring of pre-trained models	Because the health system did not build the model, responsible practice would require treating it like any other pre-trained component: demanding documentation from the vendor on training data, label choice, and subgroup performance. This is followed by continuous monitoring of its behavior after deployment, using version control to detect when vendor updates affect performance or fairness characteristics.

MANAGE 4: Documentation of risk-treatment plans

Subcategory	RMF Expectation
Manage 4.1 Monitoring post-deployment	This step includes establishing post-deployment AI monitoring processes and assessing performance metrics. The hospital network, in partnership with the vendor, must evaluate the model's reliability, validity, and bias with post-deployment TEVV tasks. Creating information sharing mechanisms with relevant AI actors about errors and near-misses in the model.[33]
Manage 4.2 Regular monitoring processes	The hospital and vendor should embed recurring activities for continual improvement into every model update. That means setting a schedule (for example, quarterly) to recalculate accuracy and fairness metrics (DP, EO, FPR/FNR by race) and review enrollment and outcome data with clinicians, equity experts, and patient representatives. When issues are found, such as widening racial gaps, improvements should target the model and surrounding workflows (e.g., strengthening clinician override guidance).
Manage 4.3 Communication Processes	Errors arising from the continuous monitoring activities must be regularly communicated to all stakeholders, particularly the affected communities. Regular documentation of all errors and recovery from the incident must be established, covering how errors were identified, the incidents resulting from the errors, whether the errors have been repaired, and the impact on affected stakeholders.[34]

In conclusion, the Optum case shows that an algorithm can appear technically effective yet produce deeply unfair real-world outcomes when its design is misaligned with the actual decision it is meant to support. As the MAP analysis demonstrates, the system was built and deployed around future healthcare spending rather than true clinical need, and this proxy choice embedded existing racial inequities directly into the model's outputs. Applying the NIST AI RMF makes it clear that responsible use of this system would require more than minor threshold adjustments. A trustworthy

[33] ibid, 54
[34] ibid, 56

redesign must begin by redefining the target from cost to one that emphasizes clinical need, embedding fairness metrics such as DP and EO into deployment decisions, and establishing continuous monitoring and feedback loops with affected stakeholders. This case showcases that in high-stakes domains like healthcare, fairness cannot be treated as a secondary optimization goal after accuracy or efficiency. It must be built into how risk is defined, measured, and managed from the start.

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